PYTHON PROGRAMMING

ACTIVITY-BASED LEARNING • CASE STUDY

Case Study 1: University Scholarship Allocation

Name: B. Bhanushri **Roll Number:** 26000865

Branch: Computer Science & Engineering (AI & ML) **Section:** CSM1

1. Problem Statement

Evaluate a scholarship applicant using three subject marks, attendance, and family annual income.

2. Rules and Assumptions

• Each subject mark must be from 0 to 100.

• Each subject must have at least 50 marks.

• Attendance must be at least 75%.

• Family annual income must not exceed Rs. 300,000.

3. Python Source Code

Copy the following code into a .py file and run it in Python.

```
name = input("Applicant name: ")
m1 = float(input("Subject 1 marks: "))
m2 = float(input("Subject 2 marks: "))
m3 = float(input("Subject 3 marks: "))
attendance = float(input("Attendance percentage: "))
income = float(input("Family annual income (Rs.): "))

if (m1 < 0 or m1 > 100 or m2 < 0 or m2 > 100 or
    m3 < 0 or m3 > 100 or attendance < 0 or
    attendance > 100 or income < 0):
    print("Invalid input.")
else:
    percentage = (m1 + m2 + m3) / 3
    print("Applicant:", name)
    print("Aggregate percentage:", round(percentage, 2))
    if (m1 >= 50 and m2 >= 50 and m3 >= 50 and
        attendance >= 75 and income <= 300000):
        print("Result: Eligible for scholarship")
    else:
        print("Result: Not eligible for scholarship")
```

4. Sample Input

```
Applicant name: Anu
Subject 1 marks: 80
Subject 2 marks: 75
Subject 3 marks: 90
Attendance percentage: 85
Family annual income (Rs.): 200000
```

5. Expected Program Output

```
Applicant: Anu
Aggregate percentage: 81.67
Result: Eligible for scholarship
```

6. Explanation

A student with any subject mark below 50, attendance below 75%, or income above Rs. 300,000 is not eligible.

7. Conclusion

This case study applies Python input, variables, arithmetic operations, conditional statements, and validation to solve a practical problem.

Case Study 2: Progressive Electricity Billing

Name: B. Bhanushri **Roll Number:** 26000865

Branch: Computer Science & Engineering (AI & ML) **Section:** CSM1

1. Problem Statement

Calculate an electricity bill using progressive unit slabs, a fixed charge, and a high-consumption surcharge.

2. Rules and Assumptions

• First 100 units: Rs. 2 per unit.

• Next 100 units: Rs. 3 per unit.

• Units above 200: Rs. 5 per unit.

• Fixed charge: Rs. 50.

• Consumption above 200 units: 5% surcharge on energy charge.

3. Python Source Code

Copy the following code into a .py file and run it in Python.

```
units = int(input("Enter electricity units: "))

if units < 0:
    print("Invalid units.")
else:
    if units <= 100:
        energy = units * 2
    elif units <= 200:
        energy = 100 * 2 + (units - 100) * 3
    else:
        energy = 100 * 2 + 100 * 3 + (units - 200) * 5

    fixed = 50
    surcharge = energy * 0.05 if units > 200 else 0
    total = energy + fixed + surcharge

    print("\n--- Electricity Bill ---")
    print("Energy charge: Rs.", round(energy, 2))
    print("Fixed charge: Rs.", fixed)
    print("Surcharge: Rs.", round(surcharge, 2))
    print("Total bill: Rs.", round(total, 2))
```

4. Sample Input

```
Enter electricity units: 250
```

5. Expected Program Output

```
--- Electricity Bill ---
Energy charge: Rs. 750
Fixed charge: Rs. 50
Surcharge: Rs. 37.5
Total bill: Rs. 837.5
```

6. Explanation

For 250 units, the energy charge is 100×2 + 100×3 + 50×5 = Rs. 750.

7. Conclusion

This case study applies Python input, variables, arithmetic operations, conditional statements, and validation to solve a practical problem.

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Case Study 3: Hospital Billing and Patient Classification

Name: B. Bhanushri **Roll Number:** 26000865

Branch: Computer Science & Engineering (AI & ML) **Section:** CSM1

1. Problem Statement

Calculate consultation, diagnostic, and treatment charges, apply a senior-patient concession, and classify the final bill.

2. Rules and Assumptions

• Patients aged 60 or above receive a 10% concession.

• Basic: final bill below Rs. 1,000.

• Standard: Rs. 1,000 to below Rs. 5,000.

• High: Rs. 5,000 or above.

• Charges and age cannot be negative.

3. Python Source Code

Copy the following code into a .py file and run it in Python.

```
age = int(input("Patient age: "))
consult = float(input("Consultation charge: Rs. "))
diagnostic = float(input("Diagnostic charge: Rs. "))
treatment = float(input("Treatment charge: Rs. "))

if age < 0 or consult < 0 or diagnostic < 0 or treatment < 0:
    print("Invalid input.")
else:
    subtotal = consult + diagnostic + treatment
    discount = subtotal * 0.10 if age >= 60 else 0
    final_bill = subtotal - discount

    if final_bill < 1000:
        category = "Basic"
    elif final_bill < 5000:
        category = "Standard"
    else:
        category = "High"

    print("\n--- Hospital Bill ---")
    print("Subtotal: Rs.", round(subtotal, 2))
    print("Concession: Rs.", round(discount, 2))
    print("Final bill: Rs.", round(final_bill, 2))
    print("Bill category:", category)
```

4. Sample Input

```
Patient age: 65
Consultation charge (Rs.): 500
Diagnostic charge (Rs.): 800
Treatment charge (Rs.): 2000
```

5. Expected Program Output

```
--- Hospital Bill ---
Subtotal: Rs. 3300
Concession: Rs. 330
```

```
Final bill: Rs. 2970
Bill category: Standard
```

6. Explanation

The senior-patient concession is calculated from the subtotal before it is subtracted from the bill.

7. Conclusion

This case study applies Python input, variables, arithmetic operations, conditional statements, and validation to solve a practical problem.

Case Study 4: Employee Payroll and Deduction Analysis

Name: B. Bhanushri **Roll Number:** 26000865

Branch: Computer Science & Engineering (AI & ML) **Section:** CSM1

1. Problem Statement

Calculate gross salary from basic pay, allowances, overtime, and incentive, then subtract a deduction to find net salary.

2. Rules and Assumptions

• HRA = 20% of basic pay.

• Other allowance = 10% of basic pay.

• Overtime = Rs. 150 per hour.

• Incentive is added to gross salary.

• Deduction = 8% of gross salary.

• All inputs must be non-negative.

3. Python Source Code

Copy the following code into a .py file and run it in Python.

```
basic = float(input("Basic pay (Rs.): "))
hours = float(input("Overtime hours: "))
incentive = float(input("Incentive (Rs.): "))

if basic < 0 or hours < 0 or incentive < 0:
    print("Invalid input.")
else:
    hra = basic * 0.20
    other = basic * 0.10
    overtime = hours * 150
    gross = basic + hra + other + overtime + incentive
    deduction = gross * 0.08
    net = gross - deduction

    print("\n--- Salary Statement ---")
    print("Basic pay: Rs.", round(basic, 2))
    print("HRA: Rs.", round(hra, 2))
    print("Other allowance: Rs.", round(other, 2))
    print("Overtime pay: Rs.", round(overtime, 2))
    print("Incentive: Rs.", round(incentive, 2))
    print("Gross salary: Rs.", round(gross, 2))
    print("Deduction: Rs.", round(deduction, 2))
    print("Net salary: Rs.", round(net, 2))
```

4. Sample Input

```
Basic pay (Rs.): 20000
Overtime hours: 2
Incentive (Rs.): 500
```

5. Expected Program Output

```
--- Salary Statement ---
```

```
Basic pay: Rs. 20000.0
HRA: Rs. 4000.0
Other allowance: Rs. 2000.0
Overtime pay: Rs. 300
Incentive: Rs. 500
Gross salary: Rs. 26800.0
Deduction: Rs. 2144.0
Net salary: Rs. 24656.0
```

6. Explanation

Net salary = gross salary – deduction.

7. Conclusion

This case study applies Python input, variables, arithmetic operations, conditional statements, and validation to solve a practical problem.

Case Study 5: Multi-Plan Telecommunications Cost Optimization

Name: B. Bhanushri **Roll Number:** 26000865

Branch: Computer Science & Engineering (AI & ML) **Section:** CSM1

1. Problem Statement

Compare three mobile plans for a given monthly data usage and budget, then recommend the cheapest plan within budget.

2. Rules and Assumptions

• Plan A: Rs. 199 includes 2 GB; extra data costs Rs. 50 per GB.

• Plan B: Rs. 299 includes 5 GB; extra data costs Rs. 30 per GB.

• Plan C: Rs. 399 includes 10 GB; extra data costs Rs. 20 per GB.

• Usage and budget must be non-negative.

3. Python Source Code

Copy the following code into a .py file and run it in Python.

```
usage = float(input("Monthly data usage (GB): "))
budget = float(input("Monthly budget (Rs.): "))

if usage < 0 or budget < 0:
    print("Invalid input.")
else:
    cost_a = 199 + max(0, usage - 2) * 50
    cost_b = 299 + max(0, usage - 5) * 30
    cost_c = 399 + max(0, usage - 10) * 20

    plans = [("Plan A", cost_a), ("Plan B", cost_b), ("Plan C", cost_c)]
    print("Plan A cost: Rs.", round(cost_a, 2))
    print("Plan B cost: Rs.", round(cost_b, 2))
    print("Plan C cost: Rs.", round(cost_c, 2))

    eligible = [(name, cost) for name, cost in plans if cost <= budget]
    if not eligible:
        print("No plan is within your budget.")
    else:
        best_name, best_cost = min(eligible, key=lambda item: item[1])
        print("Recommended plan:", best_name)
        print("Monthly cost: Rs.", round(best_cost, 2))
```

4. Sample Input

```
Monthly data usage (GB): 12
Monthly budget (Rs.): 600
```

5. Expected Program Output

```
Plan A cost: Rs. 699
Plan B cost: Rs. 509
Plan C cost: Rs. 439
Recommended plan: Plan C
Monthly cost: Rs. 439
```

6. Explanation

The program calculates excess-data charges separately for each plan before comparing prices with the budget.

7. Conclusion

This case study applies Python input, variables, arithmetic operations, conditional statements, and validation to solve a practical problem.

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Case Study 6: Railway Fare and Reservation Calculator

Name: B. Bhanushri **Roll Number:** 26000865

Branch: Computer Science & Engineering (AI & ML) **Section:** CSM1

1. Problem Statement

Calculate ticket fare from travel distance, class of travel, and passenger age, including child and senior-citizen concessions.

2. Rules and Assumptions

• Sleeper: Rs. 1.50/km.

• AC Three Tier: Rs. 2.50/km.

• AC First Class: Rs. 4.00/km.

• Children under 5 travel free.

• Passengers aged 60 or above receive a 25% concession.

• Distance and age must be non-negative; class must be 1–3.

3. Python Source Code

Copy the following code into a .py file and run it in Python.

```
distance = float(input("Distance (km): "))
print("Classes: 1. Sleeper, 2. AC 3-Tier, 3. AC First Class")
travel_class = int(input("Enter class choice (1-3): "))
age = int(input("Passenger age: "))

if distance < 0 or age < 0 or travel_class < 1 or travel_class > 3:
    print("Invalid input.")
else:
    if travel_class == 1:
        rate = 1.50
    elif travel_class == 2:
        rate = 2.50
    else:
        rate = 4.00

    base_fare = distance * rate
    if age < 5:
        final_fare = 0
    elif age >= 60:
        final_fare = base_fare * 0.75
    else:
        final_fare = base_fare

    print("\n--- Railway Ticket Summary ---")
    print("Base fare: Rs.", round(base_fare, 2))
    print("Final ticket fare: Rs.", round(final_fare, 2))
```

4. Sample Input

```
Distance (km): 800
Class choice (1-3): 2
Passenger age: 65
```

5. Expected Program Output

```
--- Railway Ticket Summary ---
```

```
Base fare: Rs. 2000
Final ticket fare: Rs. 1500
```

6. Explanation

An 800 km AC Three Tier trip has a base fare of Rs. 2,000; a 25% senior concession reduces it to Rs. 1,500.

7. Conclusion

This case study applies Python input, variables, arithmetic operations, conditional statements, and validation to solve a practical problem.

Case Study 7: Retail Billing with Multiple Discount Rules

Name: B. Bhanushri **Roll Number:** 26000865

Branch: Computer Science & Engineering (AI & ML) **Section:** CSM1

1. Problem Statement

Calculate the total purchase amount and apply a high-value purchase discount and a premium membership discount.

2. Rules and Assumptions

• Prices and quantities must be non-negative.

• Subtotal of Rs. 5,000 or more receives 5% discount.

• Premium members receive an additional 10% discount.

3. Python Source Code

Copy the following code into a .py file and run it in Python.

```
p1 = float(input("Item 1 price: "))
q1 = int(input("Item 1 quantity: "))
p2 = float(input("Item 2 price: "))
q2 = int(input("Item 2 quantity: "))
p3 = float(input("Item 3 price: "))
q3 = int(input("Item 3 quantity: "))
is_member = input("Premium member? (yes/no): ").lower()

if p1 < 0 or q1 < 0 or p2 < 0 or q2 < 0 or p3 < 0 or q3 < 0:
    print("Invalid input.")
else:
    subtotal = (p1 * q1) + (p2 * q2) + (p3 * q3)
    discount = 0
    if subtotal >= 5000:
        discount += subtotal * 0.05
    if is_member == "yes":
        discount += subtotal * 0.10
    net_amount = subtotal - discount

    print("\n--- Retail Bill Summary ---")
    print("Subtotal: Rs.", round(subtotal, 2))
    print("Total discount: Rs.", round(discount, 2))
    print("Net payable: Rs.", round(net_amount, 2))
```

4. Sample Input

```
Item 1 price: 3000
Item 1 quantity: 1
Item 2 price: 2000
Item 2 quantity: 1
Item 3 price: 1000
Item 3 quantity: 1
Premium member? (yes/no): yes
```

5. Expected Program Output

```
--- Retail Bill Summary ---
Subtotal: Rs. 6000
Total discount: Rs. 900
Net payable: Rs. 5100
```

6. Explanation

In this example, the total discount is 5% + 10% of the subtotal = 15% of Rs. 6,000.

7. Conclusion

This case study applies Python input, variables, arithmetic operations, conditional statements, and validation to solve a practical problem.

Case Study 8: Manufacturing Cost and Profitability Analysis

Name: B. Bhanushri **Roll Number:** 26000865

Branch: Computer Science & Engineering (AI & ML) **Section:** CSM1

1. Problem Statement

Calculate total manufacturing cost, cost per unit, target profit, and selling price per unit.

2. Rules and Assumptions

• Material, labor, and overhead costs must be non-negative.

• Units produced must be greater than zero.

• Target profit is 20% of total manufacturing cost.

3. Python Source Code

Copy the following code into a .py file and run it in Python.

```
material = float(input("Material cost: Rs. "))
labor = float(input("Labor cost: Rs. "))
overhead = float(input("Overhead cost: Rs. "))
units = int(input("Units produced: "))

if material < 0 or labor < 0 or overhead < 0 or units <= 0:
    print("Invalid input.")
else:
    total_cost = material + labor + overhead
    cost_per_unit = total_cost / units
    profit = total_cost * 0.20
    total_revenue = total_cost + profit
    selling_price = total_revenue / units

    print("\n--- Manufacturing Summary ---")
    print("Total manufacturing cost: Rs.", round(total_cost, 2))
    print("Cost per unit: Rs.", round(cost_per_unit, 2))
    print("Target profit (20%): Rs.", round(profit, 2))
    print("Selling price per unit: Rs.", round(selling_price, 2))
```

4. Sample Input

```
Material cost (Rs.): 50000
Labor cost (Rs.): 30000
Overhead cost (Rs.): 20000
Units produced: 1000
```

5. Expected Program Output

```
--- Manufacturing Summary ---
Total manufacturing cost: Rs. 100000
Cost per unit: Rs. 100
Target profit (20%): Rs. 20000
Selling price per unit: Rs. 120
```

6. Explanation

Selling price per unit = (total cost + 20% profit) ÷ units produced.

7. Conclusion

This case study applies Python input, variables, arithmetic operations, conditional statements, and validation to solve a practical problem.

Case Study 9: Multi-Currency Travel Budget Analysis

Name: B. Bhanushri **Roll Number:** 26000865

Branch: Computer Science & Engineering (AI & ML) **Section:** CSM1

1. Problem Statement

Convert an Indian Rupee budget into USD, EUR, or GBP and compare it with estimated expenses.

2. Rules and Assumptions

• 1 USD = Rs. 83.

• 1 EUR = Rs. 90.

• 1 GBP = Rs. 105.

• Budget and estimated expenses must be non-negative.

3. Python Source Code

Copy the following code into a .py file and run it in Python.

```
inr_budget = float(input("Enter budget in INR: "))
print("Choose currency: 1. USD, 2. EUR, 3. GBP")
choice = int(input("Enter choice (1-3): "))
expense = float(input("Estimated trip expense in foreign currency: "))

if inr_budget < 0 or expense < 0 or choice < 1 or choice > 3:
    print("Invalid input.")
else:
    if choice == 1:
        converted = inr_budget / 83
        c_name = "USD"
    elif choice == 2:
        converted = inr_budget / 90
        c_name = "EUR"
    else:
        converted = inr_budget / 105
        c_name = "GBP"

    print("\n--- Travel Budget Summary ---")
    print("Converted Budget:", round(converted, 2), c_name)
    if expense <= converted:
        print("Status: Within budget")
    else:
        print("Status: Budget exceeded")
```

4. Sample Input

```
Budget in INR: 83000
Currency choice (1 USD, 2 EUR, 3 GBP): 1
Estimated trip expense: 900
```

5. Expected Program Output

```
--- Travel Budget Summary ---
Converted Budget: 1000.0 USD
Status: Within budget
```

6. Explanation

This case study uses the fixed exchange rates specified in the source document, not live exchange rates.

7. Conclusion

This case study applies Python input, variables, arithmetic operations, conditional statements, and validation to solve a practical problem.

Case Study 10: Campus Event Financial Planning

Name: B. Bhanushri **Roll Number:** 26000865

Branch: Computer Science & Engineering (AI & ML) **Section:** CSM1

1. Problem Statement

Calculate ticket and sponsorship revenue, subtract event expenses, and report profit, loss, or break-even.

2. Rules and Assumptions

• Ticket revenue = tickets sold × price per ticket.

• Total revenue = ticket revenue + sponsorship.

• Net result = total revenue – expenses.

• All inputs must be non-negative.

3. Python Source Code

Copy the following code into a .py file and run it in Python.

```
tickets = int(input("Tickets sold: "))
price = float(input("Price per ticket: Rs. "))
sponsorship = float(input("Total sponsorship amount: Rs. "))
expenses = float(input("Total event expenses: Rs. "))

if tickets < 0 or price < 0 or sponsorship < 0 or expenses < 0:
    print("Invalid input.")
else:
    revenue = (tickets * price) + sponsorship
    net = revenue - expenses
    print("\n--- Event Financial Statement ---")
    print("Total Revenue: Rs.", round(revenue, 2))
    print("Total Expenses: Rs.", round(expenses, 2))
    if net > 0:
        print("Net Profit: Rs.", round(net, 2))
    elif net < 0:
        print("Net Loss: Rs.", round(abs(net), 2))
    else:
        print("Break-even (No profit, no loss)")
```

4. Sample Input

```
Tickets sold: 200
Price per ticket (Rs.): 100
Sponsorship amount (Rs.): 15000
Total event expenses (Rs.): 25000
```

5. Expected Program Output

```
--- Event Financial Statement ---
Total Revenue: Rs. 35000
Total Expenses: Rs. 25000
Net Profit: Rs. 10000
```

6. Explanation

The example has Rs. 35,000 total revenue and Rs. 25,000 expenses, resulting in Rs. 10,000 profit.

7. Conclusion

This case study applies Python input, variables, arithmetic operations, conditional statements, and validation to solve a practical problem.